

SolarVillage

A New Dawn For Community Managed Solar MicroGrids

Full Conceptual Design Document



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1. Executive Summary

SolarVillage is an initiative to bring transparent, sustainable, and inclusive access to electricity to rural communities in Northern Nigeria. The challenge is not just technological. Solar panels and mini-grids are increasingly deployed, yet financial systems around repayment remain broken. Households often travel long distances to banks or solar offices to make payments. Middlemen add costs or siphon funds. Contractors cannot forecast revenue and hesitate to expand. Donors and philanthropists are willing to invest but lack visibility into whether their funds are being repaid or producing measurable outcomes. Communities are left with inconsistent service, and regulators lack reliable data to guide policy.

The SolarVillage platform addresses this by creating a seamless flow from household payment to energy delivery. A villager transfers money from their bank account or mobile wallet, either through a mobile-first web application or through a USSD menu accessible on feature phones via Africa Talks. The payment is instantly converted into Compliant Naira (cNGN), Nigeria's first regulator-approved stablecoin, and logged transparently on-chain. Depending on specific technology, the system then takes one of two possible routes:

- **Online meters (main option):** The system remotely turns on the meter, and starts checking the usage daily. Funds are then used on a daily basis until the balance hits zero, at which point the meter is automatically switched off.
- **Off-line meters (fallback option):** The system generates a time-bound activation code. When entered into the household's solar meter, this code unlocks electricity for a fixed period, usually several days or weeks depending on the payment size.

The entire process is simple, familiar, and immediate, mirroring the way Nigerians already buy airtime.

The choice of Compliant Naira as the settlement layer is central to SolarVillage's design. Unlike cash or informal ledgers, every payment and disbursement is immutably recorded, visible to regulators, donors, and contractors alike. This ensures that contractors can trust they will be paid, donors can verify that repayments are happening, and communities know they are being charged fairly. UNDP and other development partners will oversee the platform during its early phases, providing legitimacy and independent reporting.

SolarVillage benefits each stakeholder in concrete ways. Communities gain affordable and reliable energy without the burden of travel or intermediaries. Contractors enjoy predictable maintenance revenue streams that allow them to expand service. Donors and philanthropists see measurable impact with full transparency. Regulators observe a credible, responsible use of a regulated stablecoin, aligned with national energy access strategies. The Compliant Naira team gains a flagship use case that proves its token can support socially valuable applications beyond trading. Banks and mobile money operators gain transaction, volume and new rural customers.

The system architecture is straightforward. A payments orchestrator integrates with local banks, mobile money services, and the Compliant Naira ledger. Custody wallets hold funds until they are consumed (through either daily consumption monitoring and/or activation codes issued through the meter service API). Solar meters validate funds available and deliver electricity. A monitoring dashboard aggregates financial and energy data, allowing UNDP and philanthropists to view real-time metrics such as households served, repayment rates, and carbon emissions avoided.

The plan unfolds in three phases. In the first six months, SolarVillage will pilot in three to five villages, with direct UNDP oversight and donor seed capital. The goal is to validate the technology, train households and contractors, and demonstrate repayment reliability. In the following year, the platform will expand to twenty or more villages, onboard additional contractors, and introduce village-level DAOs to give communities a role in governance. Donors will be able to track village-level impact through dedicated dashboards. By the third year, the platform will reach national scale, powering tens of thousands of households, generating carbon credits, and integrating with microfinance and grid connections.

The financial model is deliberately modest and sustainable. Households pay affordable fees for energy top-ups. A very small platform fee is retained to support maintenance and compliance of the app, and another portion is collected and forwarded to local microgrid maintenance contractors. Contractors pay nominal service fees in return for reliable revenue streams. Donor funds are recycled as repayments flow back, multiplying the impact of each contribution. Carbon credits and microfinance opportunities add further layers of sustainability and incentive alignment. All transactions are visible on-chain, eliminating opacity and corruption.



Figure 1: Website concept

Risks remain, from adoption hurdles to technical failures or regulatory changes. These are mitigated through community engagement, phased rollouts, redundancy in infrastructure, and continuous dialogue with the Central Bank of Nigeria and the Rural Electrification Agency. Transparent reporting and UNDP oversight further strengthen trust. The system is inclusive by design, ensuring both smartphone and feature phone users are supported, leaving no one behind.

Beyond its core purpose, SolarVillage offers powerful expansion opportunities. The model can scale to other West African countries facing the same energy and payment challenges. Compliant Naira can be extended to the Cardano ecosystem, connecting to global pools of impact finance. Solar meter data can feed into carbon credit registries, creating new income streams for communities. Repayment histories can serve as informal credit scores, unlocking access to loans for households and small businesses. Over time, the platform could evolve into a broader rural services wallet, handling payments for water, school fees, or health insurance.

SolarVillage matters now because it aligns an urgent social need with a regulatory window of opportunity. Northern Nigeria has enormous unmet demand for electricity. Stablecoins like Compliant Naira are newly approved and seeking meaningful applications. USSD APIs make it possible to serve even the most remote, offline communities. Philanthropists and development agencies are eager for models that combine innovation with measurable impact. If brought together, these factors can create a virtuous cycle of energy access, trust, and growth.

The call to action is clear. UNDP is invited to provide oversight and legitimacy, positioning SolarVillage as a flagship demonstration of how blockchain and stablecoins can serve development goals. Philanthropists and impact investors are invited to finance pilots and early scale-up, knowing that every naira can be tracked, measured, and recycled. Contractors and solar providers are invited to join the network, benefiting from guaranteed payments and expanding their footprint. Banks and mobile money operators are invited to partner as gateways to new customer bases. Government agencies are encouraged to integrate SolarVillage into rural electrification strategies as a model for innovation and inclusion.

Join us in building a future where every household, no matter how remote, can access affordable and sustainable energy.

SolarVillage is ready to show that with the right mix of technology, finance, and governance, rural electrification in Northern Nigeria can shift from fragmented, donor-dependent projects to a transparent, scalable, and sustainable system. The platform offers not just light, but trust, opportunity, and dignity to communities that have waited too long for reliable energy.

2. Background & Problem Statement

Energy Access in Northern Nigeria

Northern Nigeria is among the most energy-poor regions in Africa. The majority of households, schools, and clinics lack access to reliable electricity, either because the national grid has never been extended to their communities or because service is unreliable and intermittent. Many communities depend on kerosene lanterns, wood, or diesel generators, which are costly, polluting, and unsafe. This energy poverty constrains daily life: children cannot study after dark, small businesses cannot refrigerate goods or use modern machinery, and healthcare facilities cannot safely store vaccines or provide reliable lighting.

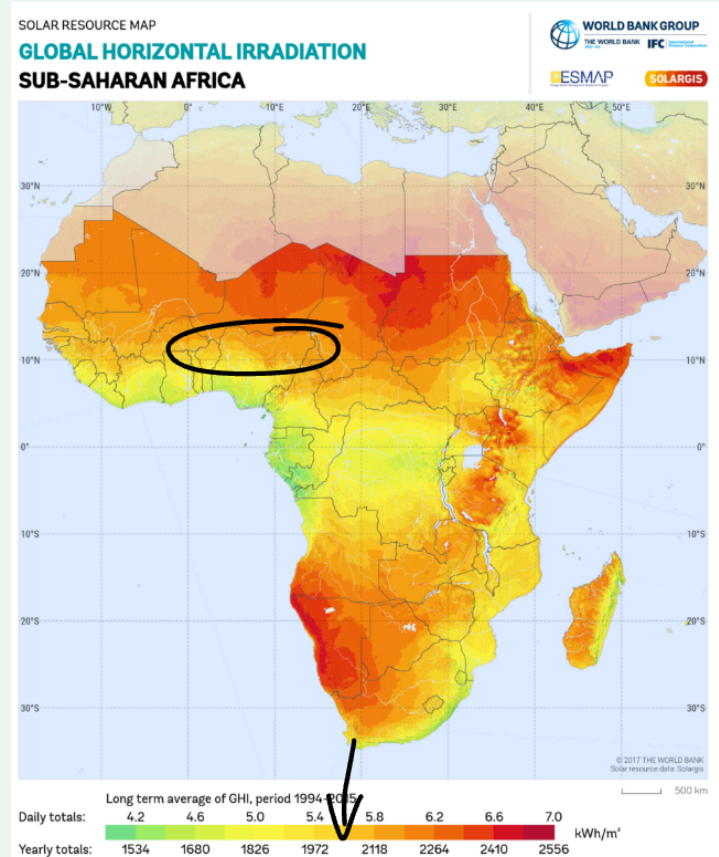
The Rise of Solar

In recent years, solar home systems and village-scale mini-grids have offered hope. NGOs, development agencies, and contractors have begun deploying solar installations, and there is increasing awareness of solar as a sustainable solution. However, the limiting factor is not the technology itself, but how payments are handled. Without efficient and transparent mechanisms for collecting user payments and recycling those payments back into maintenance and scale-up, many solar programs stall after initial deployment.

Africa is often considered and referred to as the "Sun continent" or the continent where the Sun's influence is the greatest. According to the "World Sunshine Map", Africa receives many more hours of bright sunshine during the course of the year than any other continent of the Earth, and many of the sunniest countries on the planet are in Africa.

Specifically, solar installations in Northern Nigeria can expect to produce >2,000 kWh per square meter per year.

Despite its immense solar power potential, Africa continues to lag behind other continents when it comes to building up grid and off-grid solar capacity, in part due to a lack of investment.



Payment and Trust Challenges

Today, most solar repayment systems face five critical problems:

1. **Inconvenience for users** – People must travel to distant banks or solar offices to top up their accounts, perhaps losing a day's wages just to make a payment.
2. **Costly middlemen** – Agents and intermediaries insert themselves into the process, taking commissions, sometimes even diverting funds, and eroding trust.
3. **Inconsistent repayment flows** – Contractors cannot forecast revenues, making it risky to expand into new villages.
4. **Exclusion of low-tech users** – Smartphone apps work in cities, but most rural households only have basic feature phones without internet access, leaving them cut off from digital payment systems.
5. **Patchwork of inefficient solutions** – Without a standard implementation model, local projects must invent everything themselves for every new MicroGrid project.



Philanthropists and impact investors, meanwhile, want to finance solar expansion but lack visibility into repayment performance or assurance that their funds are used effectively. Development partners like UNDP want measurable results and transparent tracking, but today's systems offer little of either. This lack of trust creates a vicious cycle: with weak repayment visibility, contractors hesitate to expand, investors hesitate to finance, and communities remain under-served.

Regulatory & Financial Context

Nigeria's financial system is evolving, with mobile money adoption rising. Yet interoperability is limited, and last-mile access in rural areas remains patchy. The **Compliant Naira (cNGN)** stablecoin—already approved by Nigerian regulators—offers a new path. It combines the familiarity of the naira with the transparency and auditability of blockchain settlement. What's missing is a flagship use case that demonstrates its real-world impact in critical sectors like energy access.

Why Now

We stand at an inflection point where several forces converge:

- **Massive unmet demand** for affordable, reliable electricity.
- **Existing solar infrastructure** awaiting scalable repayment systems.
- **Regulatory readiness** with cNGN stablecoin already cleared for use.
- **Technical enablers** like Africa Talks' USSD APIs that make blockchain transactions accessible even on feature phones.

Problem Statement

Despite the growth of solar deployment, the **absence of transparent, efficient, and inclusive payment systems continues to undermine the sustainability of rural solar projects**.

Communities remain energy-poor, contractors face uncertain revenues, philanthropists lack transparency, and regulators cannot measure success. Without intervention, Northern Nigeria risks perpetuating cycles of under-funded, unreliable solar access.

SolarVillage is designed to fill this gap: it combines compliant stablecoin payments, mobile-first and USSD interfaces, and transparent monitoring to create a sustainable, trusted model for powering Northern Nigeria.

3. Solution Overview

Concept in Brief

SolarVillage is a digital payments and management platform that makes solar energy access simple, transparent, and sustainable for rural communities in Northern Nigeria. The platform bridges the gap between users who need affordable power, contractors who install and maintain systems, philanthropists who provide financing, and regulators who demand accountability.

At its core, SolarVillage enables a **direct, traceable link** between a household's payment and the delivery of electricity through their solar meter. This is achieved by combining two critical innovations: **dual user interfaces** (mobile-first web app and USSD app) and **Compliant Naira (cNGN) stablecoin settlement**.

Everything You Need for Clean Energy

Our comprehensive solar solution brings reliable, affordable power to your community with modern payment and monitoring systems.



Clean Solar Energy

Harness the power of the sun with our high-efficiency solar panels designed for African conditions.



Flexible Payments

Pay as you go with mobile money integration. No upfront costs, just affordable daily rates.



Usage Monitoring

Track your power consumption in real-time and optimize your energy usage with smart insights.



Mobile App Control

Manage everything from your phone - payments, usage, support, and account settings.



Reliable Storage

Advanced battery systems ensure power availability even during cloudy days and at night.



Secure & Safe

Built-in safety features and theft protection keep your investment secure and your family safe.



Community Support

Local technicians and 24/7 customer support in your language ensure continuous service.

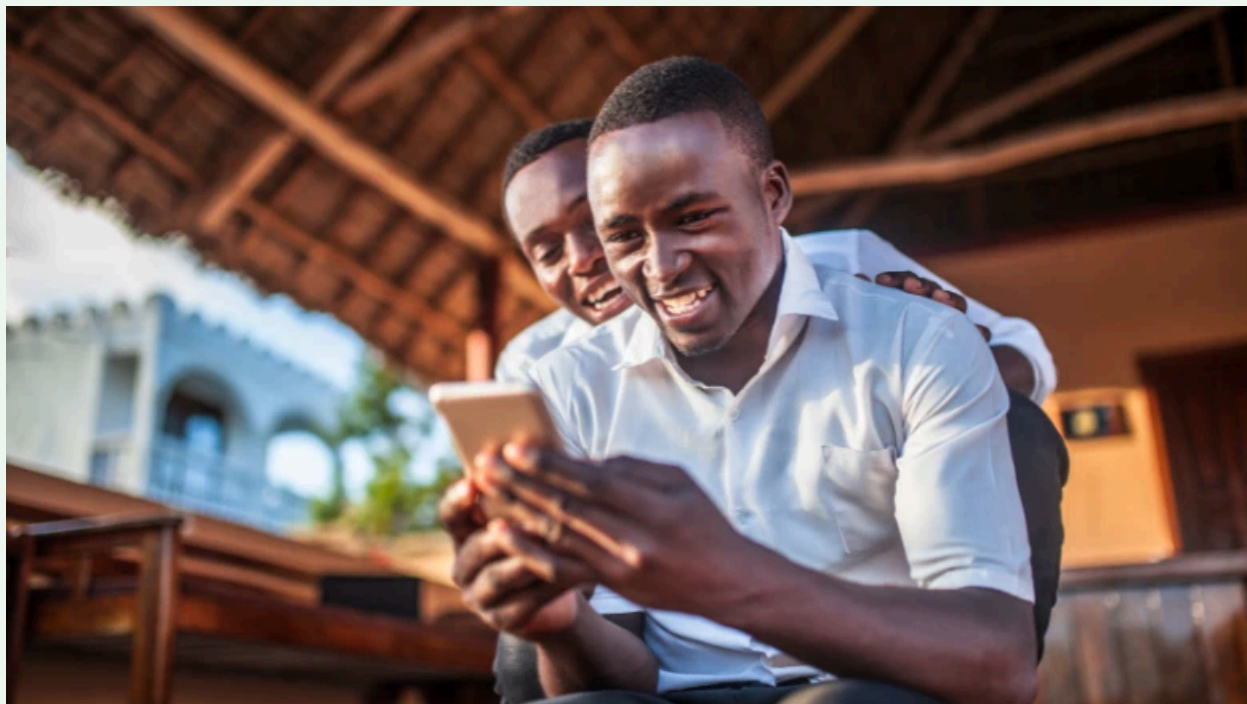


Smart Connectivity

IoT-enabled systems provide remote monitoring and automatic issue detection.

How It Works for End Users

1. A user decides to pay for electricity top-up.
2. They transfer money from their bank account or mobile wallet into the SolarVillage app.
 - Smartphone owners use the **mobile-first web app**, designed to work smoothly even on low bandwidth.
 - Feature phone owners use the **USSD menu** (via Africa Talks APIs), which requires no internet connection.
3. The payment is instantly converted into **Compliant Naira (cNGN)** stablecoin, ensuring it is recorded on a transparent ledger.
4. The system either **turns on the solar meter immediately** (default option), or generates a unique **activation code** (fallback option), valid for a specific amount of kWh
5. The user enjoys reliable energy for days, weeks or months depending on their payment amount and usage. Over time, the user may also earn credits or discounts for long term usage.
6. The meter shuts off again when either the money in the account is zero, or when the activation code expires, unless the user tops up before then.



Benefits of This Flow

- **For Users:** The process is as simple as topping up airtime; it requires no travel, no intermediaries, and provides instant results.
- **For Contractors:** Every payment is logged on-chain, reducing default risk and ensuring predictable revenues.
- **For Philanthropists & Investors:** The transparent ledger provides proof of repayment flows, enabling them to finance confidently.
- **For Regulators & UNDP:** Transaction histories can be audited in real time, ensuring accountability and measurable impact.

Dual Interface Accessibility

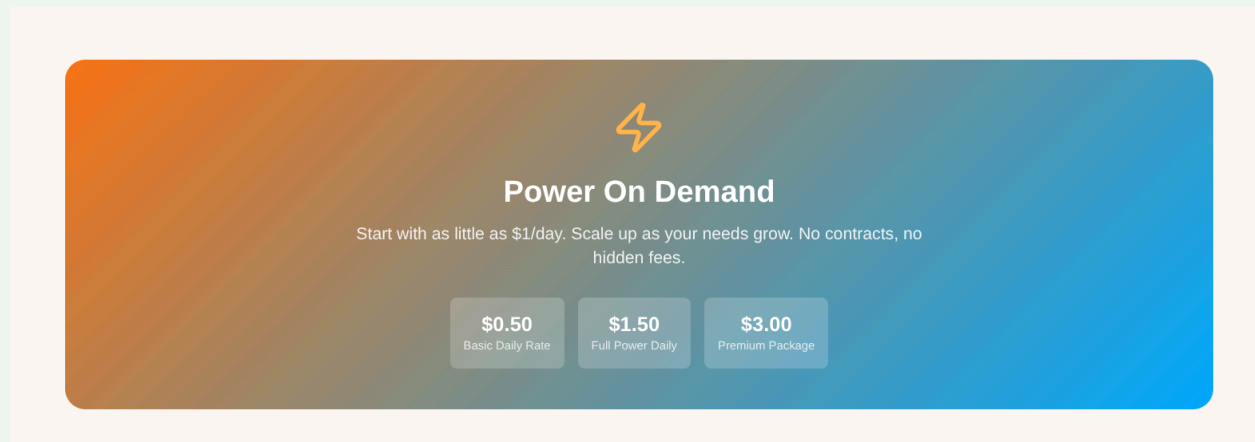
- **Mobile-first web app:** Built with responsive design, optimized for Android and low-end devices common in rural Nigeria. Features include a simple top-up flow, balance display, and QR code scanning for meter linkage.
- **USSD app:** Delivered through Africa Talks' USSD gateway, it enables anyone with a feature phone to dial a code (e.g., *123#), navigate menus in their local language, and complete payments without any internet connection required. This ensures **true**

inclusivity, covering nearly 100% of potential users.

Settlement Layer – Compliant Naira (cNGN)

All transactions settle in **Compliant Naira**, the first Nigerian naira-pegged stablecoin with regulatory approval. Key advantages:

- **Regulatory trust:** Already approved by Nigerian authorities, reducing barriers to adoption.
- **Transparency:** Every payment and disbursement is logged on-chain.
- **Efficiency:** Eliminates the need for cash-handling and manual reconciliation.
- **Scalability:** Provides a foundation for integration with DeFi and cross-border financing in the future.



Ecosystem Integration

- **Banks/Mobile Money Providers:** Serve as on-ramps, allowing users to pay via familiar local methods.
- **Solar Meter APIs:** Automatically log daily usage and/or generate time-bound activation codes linked to payments.
- **UNDP Analytics Dashboard:** Provides oversight for project partners, displaying energy access metrics, repayment rates, and impact data.



Why This Solution Stands Out

- **End-to-end simplicity:** From bank transfer to solar meter activation in a single flow.
- **Inclusion by design:** Both smartphone and feature phone users are served.
- **Trust & transparency:** Stablecoin settlement ensures all stakeholders see the same data.
- **Scalable model:** Once proven in a few villages, the system can rapidly expand across Nigeria and other regions.

Follow-on Opportunities

While the MVP is deliberately narrow and execution-focused, the architecture is then widely extensible. Several follow-on opportunities expand the impact of the platform far beyond simple payments, for example:

- Financing for PUE (Productive Use of Electricity)
- Paying for Electricity with Crops or Labor (Sarafu-style)
- Irrefutable Tracking of QAMF Indicators
- Village DAOs for Local Governance and Tariffs

Read more about these and many other follow-on opportunities in section 11. below

4. Core Stakeholders

SolarVillage is designed as a **multi-stakeholder ecosystem**, where every participant's incentives are aligned toward the shared goal of sustainable energy access. Understanding these roles and their interdependencies is key to ensuring successful adoption and scale.

4.1. Local Community Members (End Users)

- **Who they are:** Rural households, small businesses, schools, and clinics in Northern Nigeria.
- **Current pain points:** Long trips to banks or solar offices, dependence on unreliable agents, lack of control over energy costs. Lack of electric energy altogether.
- **Value from SolarVillage:**
 - Instant access to power with a simple payment → activation code → meter entry.
 - Inclusion for both smartphone and feature phone users (via web or USSD).
 - Increased trust, knowing their payments directly map to electricity access.

4.2. Contractors & Solar Installers

- **Who they are:** Local businesses and technicians who deploy and maintain solar home systems and community mini-grids.
- **Current pain points:** Irregular payments, defaults, and lack of predictable cash flow discourage them from expanding operations.
- **Value from SolarVillage:**
 - Guaranteed repayment flows logged transparently on-chain.
 - Real-time notifications of user top-ups, allowing proactive maintenance.
 - Incentives to expand installations, knowing revenues are trackable and secure.

4.3. Philanthropists & Impact Investors

- **Who they are:** Donors, NGOs, CSR programs, and social impact funds seeking measurable outcomes in energy access.
- **Current pain points:** Lack of visibility into repayment data, uncertainty about whether donations or loans are properly managed.
- **Value from SolarVillage:**
 - Transparent, auditable ledger of all repayments.
 - Ability to earmark funds for specific villages or households.
 - Measurable social impact reporting (e.g., number of households electrified, CO₂ emissions avoided).
 - Greater confidence to recycle capital into new projects.

4.4. UNDP & Development Partners

- **Who they are:** Multilateral organizations, aid agencies, and government-backed development programs.
- **Current pain points:** Difficulty demonstrating success of rural electrification initiatives due to weak monitoring and evaluation (M&E) systems.
- **Value from SolarVillage:**
 - Real-time analytics dashboard for oversight.
 - Clear KPIs: repayment rates, hours of power delivered, households served.
 - Ability to claim demonstrable success, enhancing credibility with funders and governments.
 - Replicable model that can be exported to other regions.

4.5. Compliant Naira (cNGN) Team

- **Who they are:** The company and consortium behind Nigeria's first regulator-approved stablecoin.
- **Current pain points:** Need for credible, large-scale use cases that prove stablecoins can solve real-world problems beyond crypto trading.
- **Value from SolarVillage:**
 - Flagship deployment showcasing cNGN as a tool for social impact.
 - Significant transaction volumes driving adoption and liquidity.
 - Strong positioning with regulators as a cornerstone partner in financial inclusion.

4.6. Local Banks & Mobile Money Providers

- **Who they are:** Nigerian banks, payment service providers, and telco-backed mobile money operators.
- **Current pain points:** Limited penetration in rural regions; need to prove relevance in underserved markets.
- **Value from SolarVillage:**
 - Increased transaction volume from new customer segments.
 - Seamless integration as on-ramps/off-ramps for cNGN.
 - Enhanced reputation as enablers of rural financial inclusion.

4.7. Regulators & Government Agencies

- **Who they are:** Central Bank of Nigeria, Rural Electrification Agency, and state-level energy regulators.
- **Current pain points:** Need to balance innovation with consumer protection and financial stability.
- **Value from SolarVillage:**

- Assurance that cNGN is used responsibly in a controlled environment.
 - A showcase of blockchain delivering social good, not just speculation.
 - Data-driven insights into rural energy use, helping shape future policy.
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5. System Architecture

The **SolarVillage system architecture** integrates financial infrastructure, user interfaces, and hardware in a seamless flow that prioritizes accessibility, transparency, and scalability. It ensures that payments made by users are reliably converted into electricity, while stakeholders can monitor and verify every step of the process.

Key Components

1. User Interfaces

- **Web App (Mobile-First):**
 - Accessible on smartphones with low bandwidth optimization.
 - Features: wallet balance, payment history, top-up flow, QR code support for meter pairing.
- **USSD Interface** (via Africa Talks APIs):
 - Accessible on feature phones without the internet.
 - Provides simple menu navigation (*123# style) in local languages.
 - Ensures inclusivity for users at the “last mile.”

2. Payments Orchestrator

- Core middleware that processes all payment intents.
- Validates user details, links transactions to specific meter IDs, and routes funds to the correct channels.
- Handles exceptions (failed transfers, expired codes, refunds).

3. Banking & Mobile Money Integration

- Acts as the on-ramp for users.
- Accepts funds via bank transfer, debit card, or mobile money (e.g., MTN MoMo).
- Provides instant payment confirmation to the orchestrator.

4. Compliant Naira (cNGN) Ledger

- Stablecoin smart contract approved by Nigerian regulators.
- Every transaction is recorded on-chain for transparency.
- Provides settlement and liquidity between users, contractors, and philanthropists.

5. Custody Wallets

- Safekeeping accounts where cNGN is pooled before disbursement.
- Used to manage escrow (e.g., contractor payments, subsidy disbursements).
- Transparent balances visible on-chain.

6. Meter Service API

- Connects digital payments to physical electricity meters.
- Generates prepaid activation codes based on payment value.
- Communicates with solar meter hardware to validate codes and unlock power.

7. Solar Meters

- Installed at households, schools, or mini-grids.
- Accept activation codes from users.
- Record consumption data for monitoring.

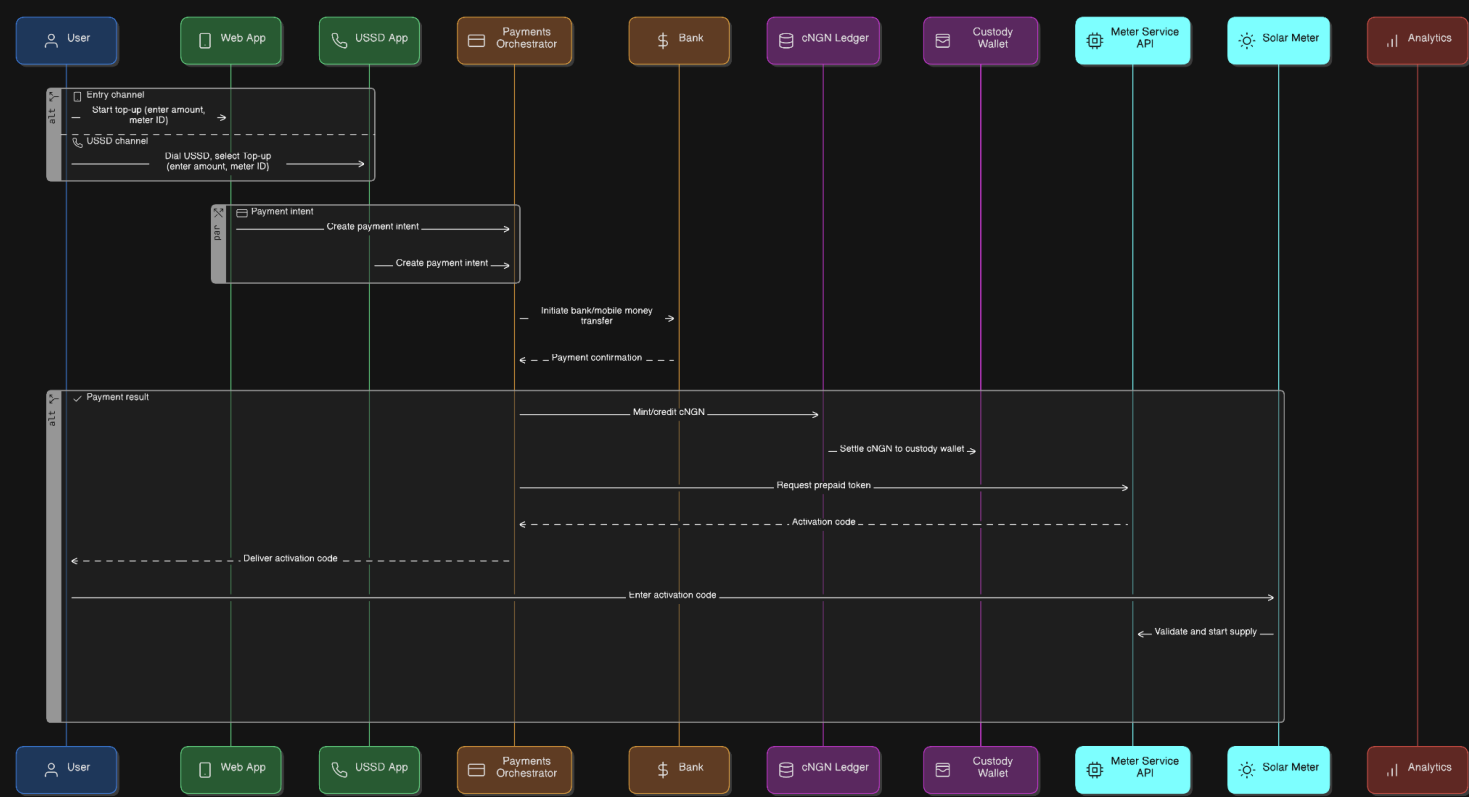
8. Analytics & Oversight Layer

- Dashboard for UNDP, regulators, and philanthropists.
 - KPIs: repayment rates, households powered, kWh delivered, CO₂ savings.
 - Aggregates both financial (on-chain) and energy (meter) data.
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Data & Payment Flow (Step by Step)

The step by step process will follow two alternative sub-flows depending on which metering infrastructure is chosen.

1. **User action:** Initiates top-up via Web App or USSD.
2. **Payments Orchestrator:** Creates payment intent and requests funds.
3. **Bank/Mobile Money:** Processes user payment, sends confirmation.
4. **cNGN Ledger:** Mints or credits equivalent cNGN stablecoin.
5. **Custody Wallet:** Holds user's cNGN until it's mapped to an activation or consumption.
6. **Meter Service API:**
 - *In case of online metering (target state)*
 - i. Communicates daily with the meter to keep it activated if funds are available.
 - ii. Collects usage statistics daily, based on 15-minute increments.
 - iii. Pushes activation to the meter as soon as new payment is received.
 - *In case of offline metering with activation codes (fallback option)*
 - i. Generates an activation code tied to the payment.
7. **User:**
 - *In case of online metering (target state)*
 - i. Enjoys electricity as soon as payment is made.
 - *In case of offline metering with activation codes (fallback option)*
 - i. Enters code into their solar meter
 - ii. Can enjoy electric power immediately.
8. **Solar Meter:**
 - *In case of online metering (target state)*
 - i. Turns off when no funds left
 - ii. Turns off when no communication
 - iii. Reports usage statistics daily
 - iv. Turns on when it receives an activation call.
 - *In case of offline metering with activation codes (fallback option)*
 - i. Validates code and enables electricity for a set duration
9. **Analytics Layer:** Logs transaction and energy data for reporting.



6. Benefits Analysis

SolarVillage is designed as a win-win ecosystem. Each stakeholder group receives tangible benefits that incentivize participation and create a reinforcing cycle of adoption, trust, and expansion.

6.1. Local Community Members (End Users)

The key reason we are passionate about this project is that it ensures access to affordable, reliable, sustainable and modern energy for all, which also happens to be one of the [17 global Sustainable Development Goals](#) (SDGs) of the United Nations.

- **Convenience:** No need to travel to banks or solar offices; payments can be made instantly from home.
- **Inclusivity:** Both smartphone users (via web app) and feature phone users (via USSD) are supported.
- **Energy Security:** Payments translate directly into electricity, providing reliable and predictable power.
- **Cost Savings:** Reduced reliance on kerosene or diesel lowers household expenses and health risks.
- **Empowerment:** Families and small businesses gain control over their energy use, aligning payments with consumption.



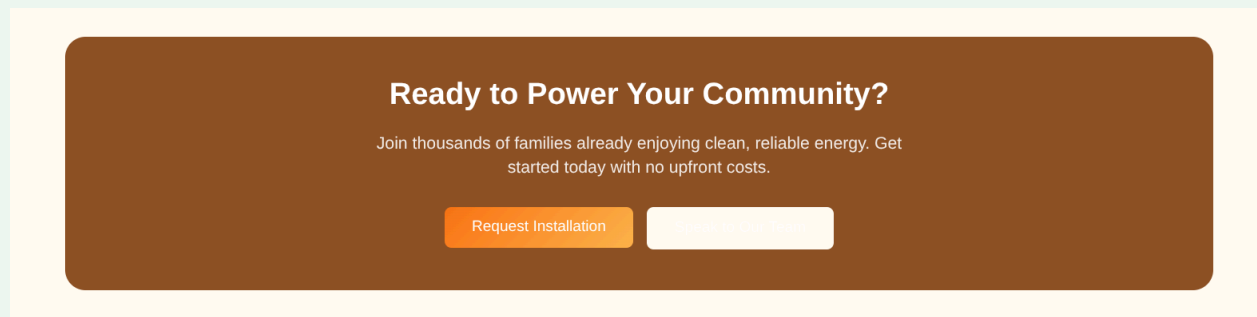
6.2. Maintenance Contractors

- **Reliable Cash Flow:** Payments are logged on-chain, reducing defaults and ensuring predictable revenue streams.
- **Transparency:** Contractors can see payment data in real time, helping with forecasting and planning.
- **Business Growth:** More reliable repayment cycles encourage expansion into new villages.
- **Reduced Risk:** No need to rely on informal intermediaries or cash collection, lowering fraud exposure.
- **Reputation Enhancement:** Contractors seen as trustworthy partners backed by a transparent system.

6.3. Philanthropists & Impact Investors

- **Proof of Impact:** Transparent, auditable repayment flows and energy metrics ensure accountability.

- **Capital Recycling:** Funds can be reinvested once repaid, multiplying impact over time.
- **Targeted Deployment:** Ability to earmark financing for specific villages, schools, or clinics.
- **Impact Reporting:** Detailed dashboards (households served, CO₂ avoided, hours of electricity delivered).
- **Reliable Repayments:** Loan repayments are logged on-chain, reducing defaults and ensuring predictable repayment streams.
- **Confidence:** Reduced risk of funds misuse, encouraging larger commitments.



6.4. UNDP & Development Partners

- **Oversight & Monitoring:** Real-time data dashboards provide visibility into performance.
- **Clear Metrics:** KPIs like repayment rates, electrification rates, and social outcomes are tracked automatically.
- **Replication:** Once proven, the model can be scaled nationally or replicated in other countries.
- **Policy Alignment:** Supports Nigeria's commitments to SDG7 (Affordable and Clean Energy) and climate targets.
- **Demonstrable Success:** SolarVillage becomes a flagship program UNDP can showcase globally.

6.5. Compliant Naira (cNGN) Team

- **Flagship Use Case:** A socially impactful deployment that demonstrates the utility of cNGN beyond trading.
- **Transaction Volume:** Significant on-chain usage that strengthens liquidity and adoption.
- **Regulatory Validation:** Reinforces cNGN's credibility with Nigerian regulators and financial institutions.
- **Ecosystem Growth:** Opens pathways to connect with DeFi, remittances, and microfinance.

6.6. Local Banks & Mobile Money Providers

- **New Customer Base:** Rural communities become active participants in digital payments.
- **Increased Volume:** Regular energy top-ups drive transaction activity.
- **Interoperability:** Banks become key on-ramps and off-ramps for cNGN.
- **Positive Branding:** Banks are seen as enablers of rural development and financial inclusion.

6.7. Regulators & Government Agencies

- **Accountability:** Assurance that approved stablecoins are being used responsibly.
 - **Policy Data:** Insights into rural electricity usage patterns inform energy planning.
 - **Social Stability:** Reduced reliance on unsafe fuels improves public health and lowers household financial strain.
 - **Proof of Innovation:** Nigeria can position itself as a leader in combining fintech and clean energy.
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7. Implementation Plan

The SolarVillage rollout will be phased to minimize risk, ensure community adoption, and establish a replicable model. Each phase is designed to build confidence among stakeholders while laying the groundwork for scale.



Phase 1: Pilot Deployment (0–6 months)

Objectives: Validate technology, demonstrate impact, and build trust with communities and partners.

- **Site Selection:** Identify 3–5 pilot villages in Northern Nigeria with unmet electricity needs and existing or planned solar installations.
- **Community Engagement:** Conduct sensitization workshops to introduce SolarVillage, explain benefits, and train households on using web and USSD interfaces.
- **System Setup:**
 - Deploy mobile-first web app and USSD app.
 - Integrate with Africa Talks for USSD and with selected local banks/mobile money operators.
 - Connect to the Compliant Naira (cNGN) ledger for settlement.

- Configure the meter service API with participating contractors.
- **Pilot Funding:** Secure blended financing (philanthropic grants + contractor co-investment).
- **Monitoring:** Establish a UNDP-led oversight dashboard for real-time tracking of payments and electricity activation.
- **KPIs:** Number of households onboarded, repayment rate, user satisfaction, contractor payout timeliness.



Phase 2: Regional Scale-Up (6–18 months)

Objectives: Expand reach, attract philanthropic capital, and prove financial sustainability.

- **Village Expansion:** Scale to 20+ villages across 2–3 states in Northern Nigeria.
- **Partnerships:** Formalize collaborations with regional banks, telcos, and mobile money providers for deeper on-ramp/off-ramp coverage.
- **Capacity Building:** Train additional contractors and community energy managers.
- **Philanthropy Integration:** Launch donor dashboard to allow philanthropists to track village-specific results.
- **Platform Enhancements:** Add multilingual support, QR-based payments, and offline balance notifications.

- **Governance Experiments:** Pilot village-level DAOs for shared solar grids.
- **KPIs:** 5,000 households powered, >90% repayment rates, measurable CO₂ reductions, first philanthropic capital recycled.

Phase 3: National Rollout & Ecosystem Expansion (18–36 months)

Objectives: Position SolarVillage as Nigeria's leading transparent rural electrification platform.

- **Geographic Reach:** Expand to hundreds of villages across multiple states.
- **Ecosystem Services:**
 - Enable microfinance products using repayment histories (credit scoring).
 - Integrate carbon credit issuance for global climate markets.
 - Connect some solar installations to the national grid for hybrid operations.
- **Advanced Governance:** Scale village DAOs with local councils overseeing shared micro-grids.
- **Philanthropy at Scale:** Introduce pooled donor funds for rapid expansion.
- **Policy Engagement:** Work with federal agencies to integrate SolarVillage into national energy access strategies.
- **KPIs:** 50,000+ households electrified, \$10M+ in cNGN transaction volume, multiple donors active, government endorsement secured.



Cross-Phase Enablers

- **Training & Support:** Continuous training of users and contractors to ensure adoption.
 - **Monitoring & Evaluation:** Independent audits and UNDP dashboards at each phase.
 - **Community Feedback Loops:** Incorporate user feedback to improve usability and trust.
 - **Scalable Infrastructure:** Ensure the architecture can handle exponential growth without compromising performance.
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8. Governance & Transparency

The credibility and long-term sustainability of SolarVillage depend on robust governance and transparent operations. The system is deliberately designed so that **all stakeholders can monitor performance, verify payments, and participate in decision-making** at the right level.

UNDP Oversight

- **Role:** Act as the primary independent oversight body during the initial phases.
- **Functions:**
 - Operate and maintain the monitoring dashboard.
 - Verify financial and energy delivery data.
 - Publish quarterly progress reports to donors, regulators, and community stakeholders.
 - Mediate disputes between contractors, communities, and funders if needed.
- **Impact:** Provides confidence to philanthropists and regulators that the program is accountable and results-driven.

Transparent On-Chain Ledger

- **Compliant Naira (cNGN)** ensures every payment is recorded on-chain.
- **Benefits:**
 - Immutable records of household top-ups, contractor payouts, and donor contributions.
 - Real-time tracking of fund flows for philanthropists and development partners.
 - Audit-ready data for regulators and independent evaluators.

Community-Level Governance (Village DAOs)

- **Introduction:** Beginning in Phase 2, SolarVillage will pilot **Village DAOs** (Decentralized Autonomous Organizations) as a way to self-govern their solar microgrid.
- **Functions:**
 - Represent the collective interests of households in a village.
 - Participate in decisions on local tariffs, maintenance schedules, and dispute resolution.
 - Manage small community funds (e.g., subsidy pools or emergency credits).
- **Benefits:** Empowers communities to take ownership of their energy infrastructure, strengthening accountability and local trust.

Contractor Accountability

- **Onboarding:** Contractors - both installers and maintainers - must meet baseline criteria (licensing, service quality standards) before joining the platform.
- **Payment Triggers:** Payouts to contractors are tied to verifiable performance metrics, such as number of meters installed, uptime of mini-grids, and completion of maintenance requests.
- **Monitoring:** Real-time payment and performance data reduces the risk of non-delivery.

Donor & Philanthropist Transparency

- **Dashboards:** Customized views allow donors to see where their funds are deployed and what outcomes they generate.
- **Impact Proof:** KPIs such as households electrified, hours of power delivered, and CO₂ saved are displayed alongside repayment flows.
- **Earmarking:** Donors can choose to fund specific villages or sectors (e.g., schools, clinics).

Regulatory Compliance

- **Central Bank of Nigeria:** Informed of all cNGN transactions linked to SolarVillage.
 - **Rural Electrification Agency:** Receives reports to align with national rural electrification goals.
 - **Audit Mechanisms:** Periodic third-party audits validate compliance with local laws and best practices.
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Why This Governance Model Works

- **Checks & Balances:** UNDP provides external oversight, communities govern locally, contractors are accountable to transparent payment rules, and donors receive verifiable reports.
 - **Trust-Building:** Transparency reduces risk of corruption, misuse of funds, or disputes.
 - **Scalability:** A governance structure that combines global oversight with local empowerment ensures the model can expand without losing legitimacy.
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9. Financial Model

SolarVillage's financial model is designed to balance **affordability for users**, **sustainability for contractors**, and **accountability for donors and regulators**. It combines micro-payments from households with donor support and small transaction fees that sustain platform operations.

Revenue Streams

Households, contractors, donors, and investors each play a role in generating predictable revenue flows that sustain the ecosystem:

1. **User Top-Ups**

Households pay directly into the system using bank transfers, mobile money, or cards. Payments are converted into Compliant Naira (cNGN) and logged on-chain. These funds go toward power activation and are split into loan repayment, contractor operations, platform fees, and—once debts are cleared—a community reserve fund.

2. **Transaction Fees**

A small fee (e.g., 1–2%) is applied to each household top-up. This covers platform maintenance, API integrations, and compliance costs. The fee is kept low to ensure energy remains affordable..

3. **Contractor Service Fees**

Contractors pay a nominal subscription or percentage fee to access the SolarVillage ecosystem. This is offset by reliable revenue streams and reduced collection risks.

4. **Donor & Philanthropic Grants**

Grants and donations cover initial hardware, training, and subsidies. Donors can earmark funds for specific communities, schools, or clinics, with full transparency of impact.

5. **Impact Investors**

Impact-focused investors provide loans or blended capital, which are repaid from user top-ups. Transparent repayment records reduce risk and make capital recycling possible.

Together, these revenue streams ensure that SolarVillage remains both affordable to users and sustainable for providers.

Expenditure & Cost Structure

On the expenditure side, costs are distributed across several categories:

- **Technology & Platform:** Development, hosting, and integrations (banks, Africa Talks USSD, cNGN ledger).
- **Community Training & Support:** Education campaigns, onboarding workshops, call center support.
- **Contractor Payments:** Disbursed directly from user top-ups, ensuring alignment between usage and revenue.
- **Monitoring & Evaluation:** UNDP dashboard maintenance and third-party audits.
- **Scaling Costs:** Expansion to new villages (logistics, hardware, new contractor onboarding).

By keeping expenditures closely tied to outcomes, the model prioritizes cost-efficiency and transparency.

Sustainability Mechanisms

To ensure long-term viability, the SolarVillage model incorporates several safeguards:

- **Recycling of Donor Funds:** Grants structured as revolving pools, replenished as repayments come in.
- **Transparent Escrow:** Contractor and donor funds held in custody wallets, visible on-chain, minimizing misuse.
- **Incentive Alignment:** Contractors expand service coverage as repayment predictability improves.
- **Cross-Subsidization:** Donors can subsidize households in extreme poverty, while better-off households pay full cost.

These mechanisms create resilience and reduce dependency on continuous outside funding.

Leverage Opportunities

Finally, SolarVillage's model creates opportunities for expansion and multiplier effects:

1. **Blended Finance:** Combine philanthropic seed funding with concessional loans to maximize impact per donor dollar.

2. **Carbon Credits:** Monetize emissions reductions (diesel and kerosene displaced) to generate additional revenue streams.
 3. **Credit Scoring:** Use repayment histories to unlock access to micro-loans for households and small businesses.
 4. **Ecosystem Partnerships:** Banks and telcos benefit from new customer segments, creating incentives for co-investment.
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Example Household Flow

To illustrate how the model works at a household level:

- The user pays ₦2,000 via mobile money.
- ₦40 (2%) retained as platform fee.
- ₦1,960 credited toward solar activation.
- Activation code delivered, unlocking 7 days of electricity.
- Transaction recorded on-chain for auditing.

This simple example shows how small, routine payments flow transparently through the system while supporting all stakeholders.

10. Risk Analysis

While SolarVillage presents a transformative model for rural electrification, there are risks that must be anticipated and proactively managed. The following categories outline key risks, their potential impact, and strategies for mitigation.

10.1. Adoption Risks

Challenge: Communities may resist new payment methods, especially those involving digital tokens, due to lack of familiarity or trust.

- **Potential Impact:** Slow uptake in early villages, undermining repayment cycles and donor confidence.
- **Mitigation:**
 - Conduct **community sensitization campaigns** with local leaders.
 - Provide **hands-on training** for both smartphone and feature phone users.
 - Leverage **USSD (Africa Talks)** to ensure the experience mirrors familiar mobile money workflows.
 - Offer **initial subsidies** (donor-funded free top-ups) to build trust.

10.2. Regulatory Risks

Challenge: Although Compliant Naira (cNGN) is already approved, shifts in regulatory policy could restrict stablecoin usage or impose new compliance requirements.

- **Potential Impact:** Interruptions in the settlement layer or reduced investor confidence.
- **Mitigation:**
 - Maintain **proactive engagement with the Central Bank of Nigeria (CBN)** and Rural Electrification Agency.
 - Ensure **audit-ready reporting** for financial flows.
 - Keep an **off-ramp contingency** (e.g., fallback to mobile money escrow) if stablecoin rails are temporarily suspended.

10.3. Technical Risks

Challenge: Failures in app performance, USSD integration, or meter APIs could disrupt service.

- **Potential Impact:** Users unable to top up or activate electricity, leading to frustration and distrust.
- **Mitigation:**
 - Build **redundant infrastructure** (e.g., multiple bank/payment partners).
 - Conduct **extensive testing** before live rollouts.
 - Maintain **24/7 contractor support lines**.
 - Use **phased rollouts** to identify bugs in small-scale pilots before scaling.

10.4. Trust & Transparency Risks

Challenge: Users, donors, or contractors may fear that funds are being misused, or that the system favors one group over another.

- **Potential Impact:** Reduced donor engagement, community pushback, or contractor withdrawal.
- **Mitigation:**
 - Ensure **all transactions are visible on-chain** through the cNGN ledger.
 - Provide **open dashboards** to donors and regulators.
 - Implement **Village DAOs** to give communities a voice in governance.
 - Regularly publish **UNDP-verified reports** to build trust.

10.5. Contractor & Maintenance Risks

Challenge: Contractors may fail to deliver on installation or maintenance commitments, undermining user experience.

- **Potential Impact:** Households lose confidence in solar solutions, affecting repayment rates.
- **Mitigation:**
 - Tie **contractor payouts to verifiable KPIs** (e.g., uptime, number of functioning meters).
 - Use **smart contracts/escrow** to release payments only upon delivery.
 - Maintain a **network of pre-qualified contractors** to step in if one fails.

10.6. Financial Sustainability Risks

Challenge: The system may become overly dependent on donor subsidies, delaying the transition to self-sustaining operations.

- **Potential Impact:** Long-term fragility once donor funds decline.
- **Mitigation:**
 - Introduce **gradual subsidy phase-outs** as repayment rates stabilize.
 - Use **carbon credits and microfinance tie-ins** to diversify revenue streams.
 - Keep **transaction fees low but sufficient** to cover operational costs.

10.7. Security Risks

Challenge: Risks of fraud, hacking, or identity theft in digital transactions.

- **Potential Impact:** Financial losses, regulatory backlash, erosion of trust.
- **Mitigation:**
 - Enforce **multi-factor authentication** and secure PIN flows.
 - Partner with **trusted custodians** for stablecoin wallets.
 - Commission **independent security audits**.
 - Maintain **incident response protocols** for rapid containment.

10.7. Counterparty Risks

Challenge: Risks of counterparties defaulting or getting hacked. This is particularly concerning when engaging with new and yet-to-be-proven entities such as *Wrapped CBDC Ltd.*

- **Potential Impact:** Financial losses, disruption of settlement flows, or a broader collapse of trust in the system.
 - **Mitigation:**
 - **Rigorous Due Diligence:** Only onboard counterparties with proven track records, audited financials, and regulatory approval.
 - **Diversification of Partners:** Avoid reliance on a single counterparty by spreading exposure across multiple providers and rails.
 - **Escrow & Safeguards:** Use custody wallets and escrow mechanisms to limit exposure in case a counterparty fails.
 - **Independent Security Audits:** Require counterparties to undergo regular third-party security reviews and share results.
 - **Contingency Protocols:** Maintain fallback settlement options (e.g., direct mobile money escrow) to ensure continuity if a counterparty collapses.
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11. Follow-on & Expansion Opportunities

Beyond its core function of facilitating transparent, inclusive payments for solar energy, **SolarVillage** creates a foundation for broader innovations. These opportunities extend the platform's impact, diversify revenue streams, and position it as a catalyst for both local development and global climate action.

11.1. Financing for PUE (Productive Use of Electricity)

Beyond household consumption, reliable electricity enables communities to power income-generating activities such as irrigation, agro-processing, or cold storage. Financing these productive-use assets strengthens the economic base of rural villages and increases repayment capacity.

- **Rationale:** Energy access is most sustainable when it directly supports livelihoods.
- **Opportunity:**
 - Use repayment data to underwrite small loans for PUE equipment.
 - Partner with equipment providers to offer bundled finance + service.
- **Impact:** Expands incomes, reduces poverty, and improves loan repayment reliability.

11.2. Paying for Electricity with Crops or Labor (Sarafu-style)

Many rural households experience periods where cash liquidity is low, but they have other forms of value: crops, livestock, or labor. By integrating voucher-style commitments similar to the Sarafu Network in Kenya, communities could pledge future harvests or work contributions as collateral for energy access. These commitments would be tokenized, recorded, and later redeemed or settled, expanding energy affordability beyond pure cash payments.

- **Rationale:** Cash shortages often block access, even when households produce other forms of value.
- **Opportunity:**
 - Tokenize commitments of crops, livestock, or labor as vouchers.
 - Redeem these commitments against energy use or community projects.
- **Impact:** Improves affordability, resilience, and inclusivity in cash-poor communities.

11.3. Irrefutable Tracking of QAMF Indicators

The asset inventory and on-chain settlement layer can be extended to track Quality Assurance and Monitoring Framework (QAMF) indicators, which development agencies require for program oversight. Metrics like uptime, household electrification, repayment rates, and maintenance records can be captured and published in tamper-proof form. This provides governments, donors, and investors with an irrefutable record of project performance, de-risking future investment and scaling.

- **Rationale:** Donors and regulators require verifiable metrics to monitor project performance.
- **Opportunity:**
 - Use on-chain records to publish uptime, repayment, and maintenance data.
 - Automate QAMF reporting with dashboards accessible to all stakeholders.
- **Impact:** Builds trust, de-risks investment, and accelerates scaling.

11.4. National Grid Integration

Where feasible, the system can be designed to hybridize with the national grid. Communities that are eventually connected to centralized infrastructure could maintain their local solar assets as backup or complementary supply. The asset inventory provides the integration point for balancing local and national flows, enabling smoother grid extension and reducing risks of stranded assets.

- **Rationale:** Many villages may eventually connect to Nigeria's national power grid.
- **Opportunity:** SolarVillage could serve as a **bridge platform**, enabling households to pay both for solar (when off-grid) and for grid electricity (once connected).
- **Impact:** Ensures continuity of financial records, simplifies user experience, and allows hybrid solar-grid billing.

11.5. Village DAOs (Decentralized Autonomous Organizations)

Beyond infrastructure, the platform can seed Village DAOs—local governance mechanisms where community members vote on tariffs, maintenance schedules, or reinvestment of surplus

revenue. DAO-based governance ensures that communities are not just passive recipients of electricity but active decision-makers, with transparent, auditable records of how local energy economies are managed.

- **Rationale:** Community ownership is key to sustainability.
- **Opportunity:** Village DAOs could:
 - Manage local energy resources democratically.
 - Vote on tariff adjustments or allocate donor subsidies.
 - Oversee contractor performance and escalate issues.
- **Impact:** Builds empowerment, local trust, and resilience against governance failure.

11.6. Carbon Credit Monetization

By replacing diesel generators with solar microgrids, each community generates measurable carbon savings. The platform's transaction records and asset inventory create a robust audit trail for carbon credit issuance. Communities and investors can monetize these credits in voluntary carbon markets, unlocking an additional revenue stream that further reduces electricity costs or funds expansion.

- **Rationale:** Replacing diesel and kerosene reduces carbon emissions.
- **Opportunity:** By linking solar meter usage data to global carbon registries:
 - Communities can earn carbon credits for verified emissions reductions.
 - Credits can be sold to corporates seeking offsets, generating new income streams.
- **Impact:** Adds long-term sustainability and makes SolarVillage attractive to climate finance investors.

11.7. Microfinance & Credit Scoring

As households and cooperatives build repayment histories through the prepaid system, this data can be transformed into community credit scores. These scores can serve as a foundation for microfinance institutions, banks, or NGOs to provide loans for education, healthcare, or productive assets. This builds a bridge from energy access to broader financial inclusion.

- **Rationale:** Millions of rural Nigerians are excluded from formal credit markets.
- **Opportunity:**
 - Use repayment histories in SolarVillage to generate **trust-based credit scores**.
 - Partner with microfinance institutions to offer small loans for agriculture, education, or business.
- **Impact:** Expands economic empowerment beyond energy access.

11.8. End-of-Life Decommissioning of Minigrid Components

Solar microgrids come with long-term sustainability challenges, including what happens when batteries, inverters, or panels reach the end of their usable life. By linking every asset into the registry at deployment, the platform can provide lifecycle traceability, ensuring components are responsibly decommissioned, recycled, or repurposed. This creates accountability and opens opportunities for circular-economy financing models.

- **Rationale:** Energy projects must ensure environmental responsibility beyond deployment.
- **Opportunity:**
 - Catalog each component at installation for lifecycle traceability.
 - Partner with recyclers and manufacturers for take-back schemes.
- **Impact:** Prevents waste, supports circular economy models, and attracts climate finance.

11.9. Compliant Naira Expansion to Cardano

While SolarVillage begins with cNGN on its home networks, expansion to other blockchains can unlock new partnerships and liquidity. Cardano, with its strong presence in Africa and emphasis on social impact, provides a natural next step. Deploying cNGN on Cardano connects SolarVillage to a global ecosystem of mission-aligned investors and DeFi tools.

- **Rationale:** Cardano has one of the strongest communities in Africa, with active programs in education, DeFi, and social impact.

- **Opportunity:** Deploying Compliant Naira (cNGN) on Cardano would:
 - Attract global impact investors already aligned with Cardano’s mission.
 - Enable access to DeFi lending, liquidity pools, and staking models.
 - Provide international visibility and credibility for cNGN as a cross-chain asset.

11.10. Integration with Other Public Services

Energy payments are only one part of a household’s financial responsibilities. Once trusted, the SolarVillage wallet can evolve into a broader utility platform that simplifies how rural families pay for essential services. By expanding beyond electricity, the system strengthens digital inclusion and builds long-term value for users.

- **Rationale:** Once established, the SolarVillage platform can process more than just energy payments.
- **Opportunity:**
 - Enable payments for water supply, community health insurance, or school fees.
 - Create bundled “utility wallets” for rural households.
- **Impact:** Strengthens digital inclusion while reducing transaction friction for essential services.

11.11. Cross-Border Applications

Challenges with energy access and loan repayment are not unique to Nigeria. Neighboring countries across West and East Africa face similar barriers. By adapting the model to other compliant stablecoins, SolarVillage can become a regional solution, fostering cross-border learning and investment.

- **Rationale:** Energy poverty and solar repayment challenges are common across West Africa.
- **Opportunity:** With Compliant Naira as the anchor, the platform could be adapted to other compliant stablecoins (e.g., Ghana cedi, Kenyan shilling).
- **Impact:** SolarVillage evolves into a **regional model** for rural energy payments.

12. Impact Metrics

To ensure accountability and demonstrate value to communities, contractors, donors, and regulators, SolarVillage will track **multi-dimensional Key Performance Indicators (KPIs)**. These metrics will be automatically collected through the platform's digital architecture (payments ledger, meter API, and analytics dashboards).

12.1. Energy Access Metrics

- **Households Electrified:** Number of unique households and businesses connected to solar through the platform.
- **Electricity Delivered (kWh):** Aggregate energy consumed by users, directly linked to payment flows.
- **Uptime:** Percentage of time solar systems remain operational and delivering power.
- **Activation Codes Issued:** Number of top-ups processed per household, measuring frequency of use.

12.2. Financial Metrics

- **Transaction Volume (cNGN):** Total value of household payments processed via Compliant Naira.
- **Repayment Rates:** Ratio of completed payments versus expected payments (important for donor confidence).
- **Contractor Payouts:** Amount disbursed to installers and operators, tracked in real time.
- **Donor Funds Recycled:** Percentage of philanthropic funds repaid and redeployed into new projects.
- **Platform Fee Revenue:** Sustainable revenue generated to cover operations.

12.3. Social Impact Metrics

- **Time Saved:** Hours saved by households no longer traveling to banks or solar offices for payments.

- **Education Outcomes:** Proxy measure — increased evening study hours for school-aged children.
- **Healthcare Outcomes:** Number of clinics and health centers with continuous refrigeration for vaccines and medicines.
- **Gender Empowerment:** Percentage of women-led households onboarded, given women often manage household energy decisions.
- **User Satisfaction:** Survey-based Net Promoter Score (NPS) and adoption feedback.

12.4. Environmental Metrics

- **CO₂ Emissions Avoided:** Tons of carbon dioxide saved by replacing kerosene lamps and diesel generators with solar energy.
- **Fuel Cost Savings:** Estimated household expenditure reduction from avoided kerosene/diesel use.
- **Carbon Credits Generated:** Verified carbon units eligible for sale on global markets.

5. Governance & Transparency Metrics

- **On-Chain Transactions Recorded:** Total number of cNGN transactions logged transparently.
- **DAO Participation:** Number of households engaged in Village DAO voting and governance.
- **Audit Frequency:** Regularity of UNDP and third-party audits.
- **Dispute Resolution Efficiency:** Average time taken to resolve contractor or user disputes.

Why These Metrics Matter

- **For Communities:** Demonstrates tangible benefits in time, cost, and quality of life.

- **For Contractors:** Provides financial predictability and growth incentives.
 - **For Donors & Investors:** Ensures confidence through verifiable impact reporting.
 - **For Regulators:** Supplies reliable data for energy policy and financial oversight.
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13. Roadmap & Timeline

The SolarVillage rollout will be staged across three years, moving from proof-of-concept pilots to regional scaling and ultimately to a national program. Each phase is designed to build confidence, attract partners, and create the infrastructure for long-term sustainability.

Year 1: Pilot & Proof of Concept (Months 0–12)

Focus: Demonstrate viability, build trust, and validate technology.

- **Pilot Villages:** Select 3–5 communities in Northern Nigeria with existing or planned solar systems.
 - **Deployment:** Launch mobile-first web app and USSD app. Integrate with Africa Talks APIs, Compliant Naira ledger, and local banks.
 - **Community Onboarding:** Run sensitization campaigns with local leaders, train users on top-ups, and onboard contractors.
 - **Monitoring Dashboard:** Deploy UNDP's oversight platform for real-time tracking.
 - **Early Donor Support:** Secure philanthropic grants to subsidize early payments and incentivize adoption.
 - **KPIs:** 500–1,000 households onboarded, >80% repayment compliance, positive user satisfaction feedback.
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Year 2: Regional Scale-Up (Months 13–24)

Focus: Expand coverage, improve features, and attract philanthropic capital.

- **Village Expansion:** Scale to 20–30 villages across at least 2 states.
- **Contractor Network:** Train 20+ additional contractors to handle installations and maintenance.
- **Feature Enhancements:** Add QR-based payment options, local language USSD menus, and offline balance notifications.

- **Village DAOs:** Pilot decentralized governance in select communities to manage tariffs and subsidies.
 - **Donor Dashboard:** Launch customized portals for philanthropists to track impact.
 - **KPIs:** 5,000 households powered, >90% repayment rates, first carbon credits issued, donor recycling of funds.
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Year 3: National Expansion & Ecosystem Integration (Months 25–36)

Focus: Achieve scale, diversify revenue streams, and embed in national energy strategy.

- **National Reach:** Expand to 100+ villages across 5–7 states.
 - **Ecosystem Integration:**
 - Enable microfinance loans using repayment histories.
 - Monetize carbon credits through verified registries.
 - Introduce hybrid billing for communities with partial national grid access.
 - **Governance Maturity:** Establish Village DAOs as standard practice in SolarVillage communities.
 - **Policy Engagement:** Collaborate with the Rural Electrification Agency and CBN to integrate SolarVillage into national frameworks.
 - **Donor Pools:** Set up pooled philanthropic funds for rapid deployment.
 - **KPIs:** 50,000+ households electrified, ₦5B+ in cNGN transactions, \$1M+ in carbon credit revenues, government endorsement as a flagship program.
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Cross-Phase Activities

- **Training & Capacity Building:** Ongoing support for users, contractors, and community leaders.

- **Monitoring & Evaluation:** Quarterly UNDP reporting and third-party audits.
 - **Technology Upgrades:** Continuous improvement of apps, APIs, and dashboards.
 - **Stakeholder Engagement:** Regular consultations with regulators, donors, and community representatives.
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14. Partnerships & Ecosystem

SolarVillage is not a standalone project—it is an ecosystem solution that requires alignment between communities, financiers, regulators, and service providers. Strategic partnerships are central to its success, ensuring technical integration, financial flows, and social legitimacy.

14.1. Africa Talks (USSD Integration)

- **Role:** Provide USSD gateway services to reach feature phone users.
- **Value:** Ensures inclusivity by extending SolarVillage access to households without smartphones or internet.
- **Engagement Model:** API integration, local language support, and scalability to thousands of concurrent sessions.

14.2. Local Banks & Mobile Money Operators

- **Role:** Serve as financial on-ramps and off-ramps for user payments.
- **Value:** Allow households to pay using familiar channels (bank transfer, debit card, mobile money).
- **Engagement Model:** Partnerships with 1–2 anchor banks and mobile money operators (e.g., MTN MoMo) during pilot, expanding later.

14.3. Compliant Naira (cNGN) Team

- **Role:** Provide the stablecoin infrastructure for settlement.
- **Value:** Deliver regulator-approved, transparent, and tamper-proof financial rails.
- **Engagement Model:** Joint branding of SolarVillage as a flagship case study, shared liquidity provisioning, and ongoing compliance support.

14.4. Contractors & Solar Installers

- **Role:** Deploy and maintain solar home systems and mini-grids.

- **Value:** Ensure reliable hardware, local presence, and user support.
- **Engagement Model:** Contractor onboarding with performance-based payouts tied to uptime and repayment rates.

14.5. UNDP (United Nations Development Programme)

- **Role:** Oversight, monitoring, and reporting to international stakeholders.
- **Value:** Provides legitimacy, accountability, and access to global development networks.
- **Engagement Model:** Independent dashboard management, quarterly reports, and program co-sponsorship.

14.6. Philanthropists & Impact Investors

- **Role:** Provide upfront capital for pilots and scale-up.
- **Value:** Enable affordable household access while earning verifiable social and environmental impact returns.
- **Engagement Model:** Dedicated donor dashboards, earmarked community funding, and transparent impact metrics.

14.7. Regulators & Government Agencies

- **Central Bank of Nigeria (CBN):** Oversight of stablecoin usage and financial flows.
- **Rural Electrification Agency (REA):** Alignment with national energy access initiatives.
- **State Governments:** Support for local deployment and community mobilization.
- **Engagement Model:** Regular compliance reporting, policy integration, and joint public-private initiatives.

14.8. Technology & Infrastructure Partners

- **Meter Manufacturers & API Providers:** Supply hardware and digital interfaces for prepaid activation codes.
- **Cloud Hosting Providers:** Ensure reliability, redundancy, and security of the platform.
- **Cybersecurity Partners:** Conduct audits and safeguard against fraud or attacks.

14.9. Community-Based Organizations (CBOs) & NGOs

- **Role:** Mobilize local trust, facilitate sensitization workshops, and provide user feedback.
- **Value:** Act as cultural bridges and ensure adoption among rural households.
- **Engagement Model:** Formal MoUs with CBOs in pilot villages, scaling through NGO networks in Year 2+.

Why This Ecosystem Works

- **Mutual Incentives:** Each partner gains directly from SolarVillage's success—contractors expand business, philanthropists see measurable impact, banks gain new customers, regulators prove policy innovation.
 - **Built-In Transparency:** On-chain settlement and UNDP oversight keep all partners accountable.
 - **Scalability:** Partnerships provide the infrastructure to move from a few pilot villages to national coverage.
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15. Conclusion & Call to Action

SolarVillage represents more than a payment app — it is a systemic solution to one of Northern Nigeria's most pressing challenges: reliable, affordable, and transparent access to energy. By combining **Compliant Naira (cNGN)**, already approved by regulators, with **dual interfaces** (web and USSD) that reach all users regardless of device, SolarVillage creates an inclusive digital bridge between households, contractors, donors, and regulators.

Why It Matters Now

- **Communities:** Millions of households remain trapped in energy poverty, reliant on kerosene and diesel.
- **Contractors:** Struggle to expand solar deployments due to unpredictable repayment.
- **Philanthropists & Donors:** Want accountability and impact metrics before committing more funds.
- **Regulators:** Need credible, real-world use cases for approved digital financial instruments.
- **Nigeria:** Faces national commitments under SDG7 (Affordable and Clean Energy) and climate goals that require innovative, scalable solutions.

SolarVillage addresses all of these challenges simultaneously. It provides **instant access to electricity** for users, **transparent cash flows** for contractors and donors, **verifiable impact metrics** for UNDP, and a **flagship adoption case** for Compliant Naira.

What Success Looks Like

- By **Year 1**, hundreds of households in pilot villages will reliably top up and activate solar electricity with a simple code.
- By **Year 2**, thousands of households across multiple states will be powered sustainably, with donors recycling funds through verifiable repayment flows.
- By **Year 3**, SolarVillage will be a national model, powering tens of thousands of households, generating carbon credits, and proving the viability of compliant stablecoins as engines of inclusive growth.

Call to Action

We are inviting **UNDP and philanthropists** to become cornerstone partners in this journey:

- **UNDP:** Provide oversight, credibility, and monitoring infrastructure to ensure SolarVillage meets development standards and becomes a replicable model.
- **Philanthropists & Impact Investors:** Fund the pilot and early scale-up phases, knowing that every naira can be tracked, measured, and recycled for maximum impact.
- **Contractors & Solar Providers:** Join the network to expand installations with guaranteed, transparent payments.
- **Banks & Mobile Money Operators:** Partner as payment gateways, gaining access to new customer bases in rural Nigeria.
- **Government Agencies:** Endorse and support SolarVillage as part of Nigeria's rural electrification and financial inclusion strategy.

Final Note

The opportunity is urgent and unprecedented: to use a **regulator-approved stablecoin**, an **accessible dual-interface app**, and **transparent on-chain tracking** to power a region that has waited too long for reliable energy.

SolarVillage is ready to demonstrate that technology, finance, and community can come together to deliver light, trust, and opportunity.

Join us in building a future where every household, no matter how remote, can access affordable and sustainable energy.

